

EEGDfus: A Conditional Diffusion Model for Fine-Grained EEG Denoising

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Abstract—Electroencephalogram (EEG) signals are vital in understanding brain activity, but their weak amplitude makes them susceptible to various artifacts. Accurate denoising of EEG data is crucial as a preprocessing step to ensure precise analysis and interpretation. In recent years, the diffusion model has garnered significant attention as a promising approach in generative modeling. This model effectively addresses the issue of over-smoothing in existing deep learning methods and thus has the potential to generate more refined denoised EEG signals. However, the generation process of the standard diffusion model is highly random, limiting its direct application to EEG denoising tasks. To address this limitation, we propose a conditional diffusion model specifically designed for EEG denoising. In this model, the standard diffusion model's denoising network is replaced by a novel dual-branch network, where noisy EEG information is used as a condition to guide the generation of corresponding clean EEG signals. This dual-branch structure leverages the complementary strengths of convolutional neural network (CNN) and Transformer architectures, integrating multi-scale features to comprehensively extract information from the signal. Extensive experiments demonstrate the remarkable performance of EEGDfus in EEG denoising. We tested it on two public datasets. Testing on two public datasets, EEGdenoiseNet and SSED, demonstrated that after denoising, the average correlation coefficient increased to 0.983 and 0.992 for EOG

artifact removal, respectively. The proposed model outperforms commonly used baseline models, setting a new state-of-the-art benchmark in the field of EEG denoising.

Index Terms—Deep learning, diffusion model, dual-branch network, EEG denoising.

I. INTRODUCTION

ELECTROENCEPHALOGRAM (EEG) is extensively utilized in both clinical applications and neuroscience research due to its high temporal resolution and direct measurement of brain activity [1], [2]. However, EEG signals are prone to various types of noise, such as muscular and ocular artifacts, and power line interference (PLI). These artifacts hinder the analysis and interpretation of EEG data [3], [4]. Hence, the development of high-performance EEG denoising algorithms becomes imperative.

Several traditional methods for EEG denoising have been developed and demonstrate promising results. These techniques encompass filter-based solutions [5], regression methods [6], blind source separation (BSS) [7], and empirical mode decomposition (EMD) [8]. Among them, BSS methods decompose the contaminated EEG signals into several independent components and achieve EEG denoising by removing the components that belong to noise. This technique includes procedures like independent component analysis (ICA), independent vector analysis (IVA) [9], and canonical correlation analysis (CCA) [10]. Advanced hybrid approaches such as EMD-ICA and EMD-CCA have been introduced, merging different strategies to harness their collective advantages, thereby enhancing EEG signal denoising efficiency. However, these methods typically require a large number of EEG channels to achieve relatively effective denoising performance. Additionally, most traditional methods rely on additional reference channels and prior knowledge from human experts, impeding the development of end-to-end precise EEG denoising approaches.

In contrast, deep learning methods have gained traction in EEG denoising tasks due to their powerful feature extraction capabilities [11]. Various deep models have been proposed for this purpose, including fully connected neural networks (fcNNs), recurrent neural networks (RNNs) [12], convolutional neural networks (CNNs) [13], and the Transformer architecture [14]. However, these autoregressive models inevitably encounter the same issue of over-smoothing in EEG denoising tasks as they do in the fields of vision and audio. [15], [16], [17]. To mitigate

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this problem, generative adversarial networks (GANs) have been introduced. GAN-based methods, such as EEGANet [18], have demonstrated promising performance, generating more accurate predictions of clean EEG signals. The introduction of assistive loss functions in GANs has indeed contributed to the improvement of denoising accuracy to a certain extent. However, it unavoidably comes at the cost of increased model training complexity [19].

Denoising Diffusion probabilistic models (DDPMs) [20] have recently emerged as a highly promising generative approach, showcasing impressive performance in various applications [21], [22], [23]. The core concept of DDPMs involves learning the decay patterns of information caused by noise and then using these learned patterns to generate clean data. The diffusion process employed in the diffusion model leverages a Markov chain to construct a probability distribution transformation model, which is a fixed process controlled by hyperparameters. This methodology overcomes the training challenges commonly observed in GANs due to the joint training of two networks and allows for a more stable and tractable training process, making DDPMs an attractive alternative for denoising tasks [24]. Additionally, diffusion models have been proven to exhibit remarkable potential in preserving high-frequency information of the predicted signals, which is crucial for generating fine-grained denoised EEG signals [24]. However, the generative process of the original DDPMs exhibits significant randomness, making them unsuitable for precise EEG denoising.

In this work, we introduce EEGDfus, a conditional diffusion model for fine-grained EEG denoising from EEG signals. To the best of our knowledge, this marks the first implementation of diffusion models for this specific task. The model incorporates two principal components: the diffusion process (q) and the denoising process (p_θ). The diffusion process involves gradually adding white noise, sampled from a Gaussian distribution, to the clean EEG signal, resulting in a sequence that transitions from a clean EEG signal to Gaussian noise. Conversely, during the denoising process, the model is trained to incrementally eliminate noise from this sequence, effectively restoring the original, clean EEG signal.

Given the necessity to generate clean EEG corresponding to noisy EEG in the denoising task, the model requires strong constraints. To address this, we designed a dual-branch denoising model that combines Transformers and CNNs, replacing the commonly used deep denoising models in the original diffusion model. This innovative model employs two branches to simultaneously extract essential local features and long-distance dependencies [25] from the original noisy EEG signals and the noise sequences at each iteration. These features are then integrated across multiple scales, which enhances the identification and separation of various frequency bands. This approach is essential for preserving key physiological information throughout the denoising process [26], [27]. This methodology ensures that the denoised outputs accurately correspond to the original noisy EEG data. Compared to traditional autoregressive methods, the proposed EEGDfus better preserves high-frequency information in the predicted signals. This preservation is crucial for generating fine-grained EEG signals and improving denoising performance.

A simple performance comparison is presented in Fig. 1, where the horizontal axis denotes the sample points and the vertical axis represents the absolute difference between the predicted and actual signals at each point. A smaller absolute difference signifies enhanced denoising accuracy, highlighting better retention of high-frequency information within the predicted signal. The results show that the GAN-based method surpasses the Transformer based autoregressive approach in preserving high-frequency details. Notably, our proposed EEGDfus model exceeds both previous methods, achieving superior performance as indicated by the minimal absolute errors in the predictions. This outcome underscores the effectiveness of EEGDfus in generating highly refined denoised EEG signals.

The main contributions of this paper are summarized as follows:

- We introduce a conditional diffusion model specifically designed to remove artifacts from EEG signals. To the best of our knowledge, this is the first application of diffusion models for EEG denoising. Compared to existing deep learning methods, the proposed model effectively avoids the over-smoothing problem, thereby significantly improving denoising performance.
- To tailor the diffusion model specifically for EEG denoising, we developed a novel dual-branch structure to replace the conventional denoising model employed in traditional diffusion models. This structure introduces noisy EEG into the denoising process, effectively extracts local and global features from the signal, and performs multi-scale feature fusion, thereby guiding the diffusion model to generate the corresponding denoised EEG.
- The effectiveness and robustness of the proposed method were demonstrated through experiments conducted on two public datasets. The average correlation coefficients (CC) achieved were 0.983 and 0.992, respectively. These high CC values signify substantial performance enhancements over existing deep learning methods for EEG denoising.

The remainder of this paper is organized as follows. Section II offers a concise review of the related work in the field. Section III details the architecture of our proposed EEGDfus model. Section IV describes the experimental settings, the materials utilized in this work, and the experimental results. Section V discusses the various performance advantages and disadvantages of the model. Finally, Section VI concludes the paper, summarizing our findings.

II. RELATED WORK

In this section, we present a concise review of research related to EEG denoising. Generally, EEG denoising methods can be divided into two main categories: traditional techniques and deep learning-based techniques.

A. Traditional Methods

Several traditional methods have been developed for EEG artifact removal, including regression, blind source separation (BSS), and empirical mode decomposition (EMD). Regression methods estimate artifact components in EEG signals based on

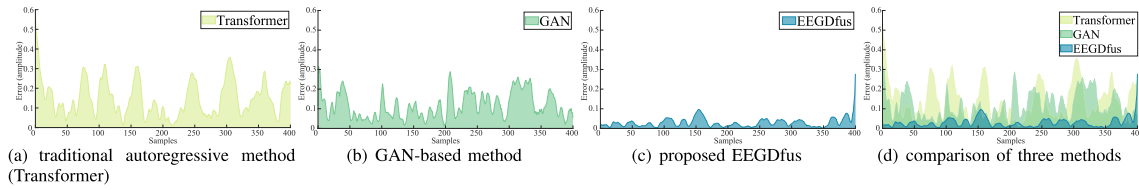


Fig. 1. Comparative analysis was conducted to evaluate the denoising performance of three different methods: (a) a traditional autoregressive method (Transformer), (b) a deep learning network based on GAN, and (c) the proposed EEGdfus network. In (d), a comprehensive comparison of the three methods is presented.

reference channels but are limited by the quality of the reference signal. BSS methods, such as ICA, CCA, and IVA, are widely used in EEG signal denoising to separate mixed signals into independent sources. ICA estimates the statistical independence of sources, effectively separating EEG signals from artifacts. CCA decomposes signals into multiple components and identifies noise components based on autocorrelation coefficients. IVA is an advanced method for separating multiple sources, particularly in complex mixing scenarios. Hybrid methods, like EMD-ICA, have further improved EEG denoising performance by combining techniques such as EMD and ICA. EMD decomposes the EEG signal into intrinsic mode functions capturing different frequency components, while ICA is then applied to the IMFs to separate the sources. However, these traditional methods generally require additional reference channels and rely heavily on prior knowledge, making it challenging to achieve end-to-end EEG denoising applications.

B. Deep Learning Based Methods

Deep learning has demonstrated significant potential in the removal of artifacts from EEG signals. A major advantage of deep learning is its capability to learn intricate mappings from input to output in an end-to-end fashion, thus eliminating the necessity for manual feature extraction. This approach enables the model to automatically identify and extract pertinent features directly from raw EEG data, thereby enhancing denoising performance.

Researchers have increasingly adopted deep learning for EEG denoising, with numerous studies demonstrating its efficacy. For instance, Nguyen et al. [28] introduced a wavelet neural network (WNN) algorithm that integrates wavelet transform (WT) and artificial neural networks to eliminate ocular artifacts without requiring EOG recordings. This approach signifies a shift from traditional to deep learning-based methods, though its performance is constrained by the limitations of WT in feature extraction. To address these limitations, Zhang et al. [12] developed a 1D-EEG denoising method leveraging convolutional neural networks (CNNs). This approach capitalizes on the ability of CNNs to automatically learn and extract significant features from raw data, effectively removing noise from EEG signals. Although the method uses a deep model to achieve end-to-end EEG denoising, CNN-based networks struggle to extract the crucial global features necessary for effective artifact removal.

To more effectively capture long-distance dependencies in EEG signals, Yi et al. [14] introduced a novel approach utilizing

the Transformer architecture. The attention mechanism in the Transformer model allows it to selectively focus on various parts of the input sequence during the encoding and decoding stages, thereby capturing long-range dependencies more effectively than traditional recurrent neural networks. Despite this advantage, autoregressive methods, including Transformers, face the issue of over-smoothing, which often results in the loss of high-frequency information in the predicted signals. This loss can impede high-precision EEG denoising. To address this issue, Sawangjai et al. [18] proposed EEGANet, a GAN-based framework for EEG denoising. By incorporating an adversarial loss, EEGANet partially mitigates the over-smoothing problem commonly encountered in traditional autoregressive models. However, this approach still has limitations in preserving high-frequency information in the generated signals, resulting in limited improvements in denoising performance.

C. Conditional Diffusion Model

The conditional diffusion model has proven to be a valuable tool in applications such as computer vision [29], [30] and text synthesis [31], [32], as well as time series processing [33], [34], where the conditioning input can be used to control the characteristics of the generated samples. This model has shown remarkable effectiveness in situations that demand fine-grained control over the generated outputs. In the context of EEG denoising, the conditional diffusion model leverages the original noisy EEG as a conditioning input to produce high-quality denoised EEG signals. This approach shows significant potential for achieving fine-grained EEG denoising.

III. METHOD

In this section, we will explore the statistical principle and optimization process behind the diffusion model. We will discuss the fundamental concepts underlying the diffusion process, including the modeling of signal degradation patterns and the generation of clean EEG signals. Additionally, we will provide a comprehensive introduction to our denoising model, outlining its essential components and their synergistic interaction to achieve precise and efficient denoising of EEG signals.

A. Denoising Diffusion Probabilistic Models

The denoising diffusion probability model (DDPM) is a type of generative model that includes two main processes: the diffusion process (q) and the denoising process (p_θ). In the diffusion

process, Gaussian noise is incrementally introduced to the clean input signal over multiple iterations, specifically T iterations. This follows a Markov process and is described by the following posterior:

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}), \quad (1)$$

where $x_1, \dots, x_T$ are the latent variables, and $q(x_{1:T} | x_0)$ is the posterior. Each step in the diffusion process is modeled by a Gaussian distribution:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t \mathbf{I}), \quad (2)$$

where $\beta_1, \dots, \beta_T$ is a sequence of hyperparameters that set the variance of the Gaussian distribution, and $\mathbf{I}$ is the unit covariance matrix with the same dimension as the input signal x_0 .

To simplify the sampling process, the re-parameterization technique is adopted. Instead of sampling directly through the Gaussian distribution $q(x_t | x_{t-1})$, the noise signal x_t at any intermediate time step is obtained directly from x_0 using the following formulas:

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_0, (1 - \alpha_t)\mathbf{I}), \quad (3)$$

$$x_t = \sqrt{\alpha_t}x_0 + \epsilon_t \sqrt{1 - \alpha_t}, \quad (4)$$

where $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{n=1}^t \alpha_n$, and $\epsilon_t \sim \mathcal{N}(0, \mathbf{I})$. Through the forward diffusion process, the complex distribution p_{complex} formed by the original data is transformed into a simple prior distribution p_{prior} with known parameters. When the number of iterations T is sufficiently large, the distribution p_{prior} is approximately a standard Gaussian distribution.

The diffusion model is capable of converting a sample from p_{prior} into one from the original data distribution p_{complex} . By using the reverse diffusion process, the model can transform a sample drawn from the Gaussian distribution $\mathcal{N}(0, \mathbf{I})$ into a sample from p_{complex} through successive sampling from $q(x_t | x_{t-1})$. The reverse diffusion process mirrors the distribution form of the forward process. When the diffusion rate β_t is sufficiently small and the number of diffusion steps is large enough, the discrete diffusion process approximates the continuous diffusion process. In this limit, the distributions of $q(x_{t-1} | x_t)$ and $q(x_t | x_{t-1})$ become Gaussian and consistent [35]. However, it is challenging to directly obtain the parameters of $q(x_{t-1} | x_t)$, so the distribution $p_{\theta}(x_{t-1} | x_t)$ is used to approximate $q(x_{t-1} | x_t)$:

$$p_{\theta}(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t)), \quad (5)$$

where μ_{θ} and Σ_{θ} represent the functions learned by the denoising model, which accepts x_t and t as parameters. Through multiple consecutive iterations, the distribution $p_{\theta}(x_0)$ can approximate the underlying clean data distribution. This can be expressed as:

$$p_{\theta}(x_0) = \int p_{\theta}(x_{0:T}) dx_{1:T}, \quad (6)$$

where p represents the joint probability distribution of x . The joint probability distribution can be derived using the conditional

probability formula:

$$p_{\theta}(x_{0:T}) = p(x_T) \prod_{t=1}^T p_{\theta}(x_{t-1} | x_t), \quad (7)$$

The deep model is trained to obtain μ_{θ} and Σ_{θ} , which can be used to determine $p_{\theta}(x_{t-1} | x_t)$, completing the denoising process.

B. Model Optimization and Add Control

Similar to other generative models, the model parameters can be estimated using maximum likelihood estimation when training data is available. Diffusion models employ maximum likelihood estimation to find the probability distribution of the Markov chain transitions during the inverse diffusion process. This means minimizing the cross-entropy between the prediction $q(x_0)$ and $p_{\theta}(x_0)$:

$$\mathcal{L} = \mathbb{E}_{x_0 \sim q(x_0)} [-\log p_{\theta}(x_0)]. \quad (8)$$

To mitigate the challenge posed by the intractable log-likelihood term in the loss function $\mathcal{L}$, we opt to maximize the lower bound on the log-likelihood. This lower bound, denoted as the variational lower bound L_{VLB} , is expressed as:

$$L_{\text{VLB}} = \mathbb{E}_{q(x_{0:T})} \left[\log \frac{q(x_{1:T} | x_0)}{p_{\theta}(x_{0:T})} \right], \quad (9)$$

The training objective can be further simplified as follows, according to [20], [36]:

$$L_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon} [\| \epsilon - \epsilon_{\theta}(\sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon, t) \|_2^2]. \quad (10)$$

The objective of EEG denoising is to derive a clean EEG signal by eliminating noise from a noisy EEG. To achieve this, additional control information is integrated into the original diffusion model. Specifically, the diffusion model is used to model the conditional distribution $p_{\theta}(x_0 | \tilde{x})$, which estimates the conditional distribution $q(x_0 | \tilde{x})$ of the denoised EEG signal. The denoising process within the conditional diffusion model can be formulated as:

$$p_{\theta}(x_{0:T} | \tilde{x}) = p(x_T) \prod_{t=1}^T p_{\theta}(x_{t-1} | x_t, \tilde{x}), \quad (11)$$

where:

$$p_{\theta}(x_{t-1} | x_t, \tilde{x}) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t | \tilde{x}), \Sigma_{\theta}(x_t, t | \tilde{x})). \quad (12)$$

Directly controlling the denoising function ϵ_{θ} with the discrete diffusion step index t may lead to degraded model performance. To address this issue, we define a noise sequence $S = 1, \sqrt{\alpha_0}, \dots, \sqrt{\alpha_T}$, where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{n=1}^t \alpha_n$. The denoising function ϵ_{θ} can then take a continuous noise level $\bar{\alpha} \sim \text{Uniform}(S_{t-1}, S_t)$ as the conditioning variable [37]. The conditional form of the original optimization problem can be expressed as follows:

$$L_{\text{condition}} = \mathbb{E}_{x_0, \tilde{x}, \bar{\alpha}, \epsilon} [\| \epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}}x_0 + \sqrt{1 - \bar{\alpha}}\epsilon, \tilde{x}, \bar{\alpha}) \|_2^2]. \quad (13)$$

The model architecture is depicted in Fig. 2, and the detailed algorithm is provided in Algorithms 1 and 2.

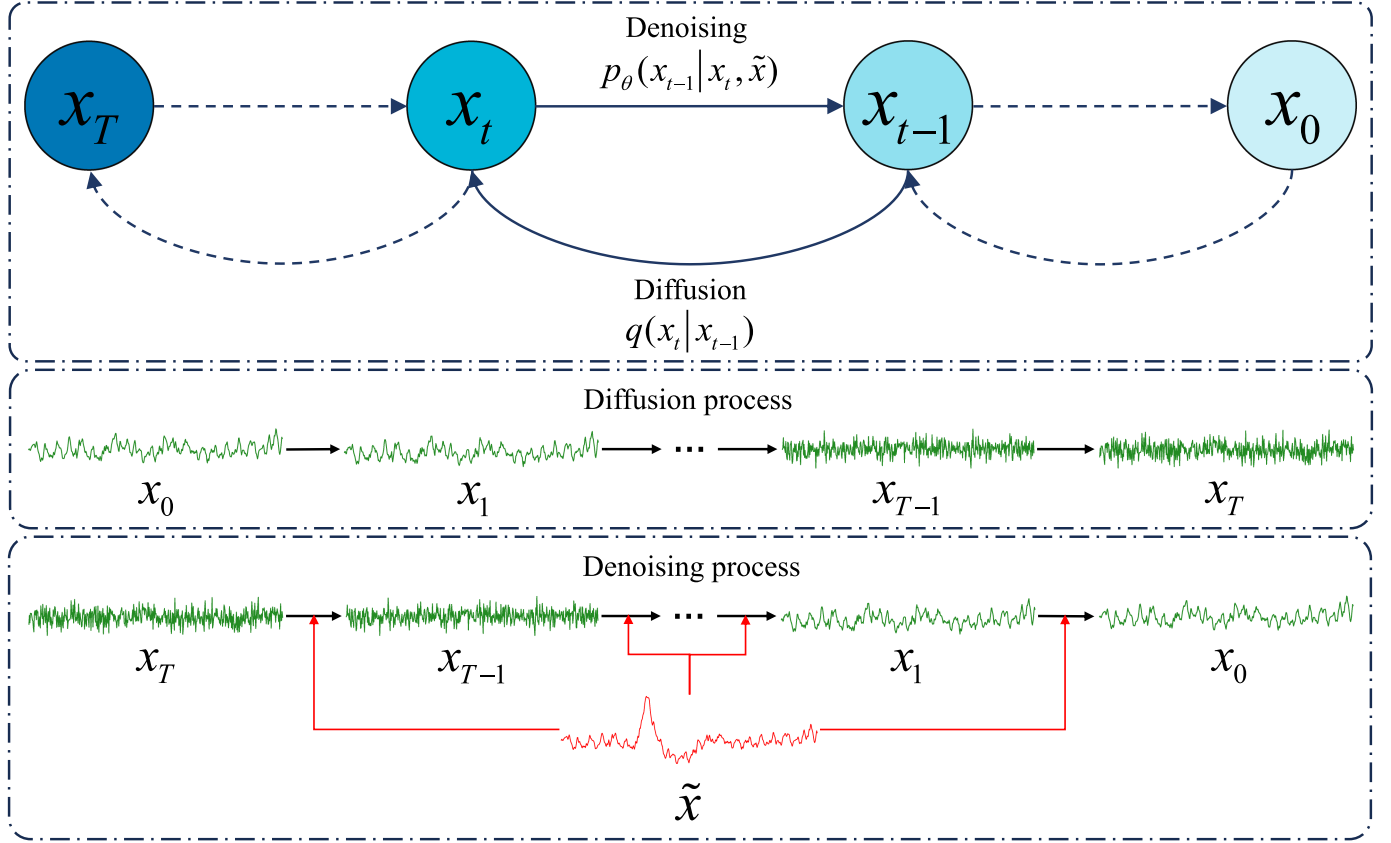


Fig. 2. Top figure illustrates the overall architecture of the proposed EEGdfus, while the two subsequent figures provide detailed depictions of the diffusion process (q) and the denoising process (p_θ), respectively.

Algorithm 1: Training.

Input: Original noisy EEG $\tilde{x}$, ground truth EEG x_0 , and noise sequence $S = \{1, \sqrt{\bar{\alpha}_0}, \dots, \sqrt{\bar{\alpha}_T}\}$
Output: Denoising function ϵ_θ

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1 repeat
2    $x_0, \tilde{x} \sim q(x_0, \tilde{x})$ ;
3    $\bar{\alpha} \sim \text{Uniform}(S_{t-1}, S_t)$ ;
4    $\epsilon \sim \mathcal{N}(0, I)$ ;
5   Take a gradient descent step on;
6    $\nabla_\theta \| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}}x_0 + \sqrt{1 - \bar{\alpha}}\epsilon, \tilde{x}, \bar{\alpha}) \|_2^2$ ;
7 until converged;
```

Algorithm 2: Inference.

Input: Denoising function ϵ_θ , noisy EEG signal $\tilde{x}$
Output: Denoised eeg signal x_0

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1  $y_T \sim \mathcal{N}(0, I)$ ;
2 for  $t = T, \dots, 1$  do
3    $z \sim \mathcal{N}(0, I)$ ;
4    $y_{t-1} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left( y_t - \frac{1 - \bar{\alpha}_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(y_t, \tilde{x}, \sqrt{\bar{\alpha}_t}) \right)$ ;
5   if  $t > 1$  then
6      $y_{t-1} = y_{t-1} + \sigma_t z$ 
7   end
8 end
9 return  $y_0$ ;
```

C. Denoising Model Structure

To adapt the diffusion model for EEG denoising, we developed a novel dual-branch denoising network that integrates a convolutional neural network (CNN) and a Transformer structure within each branch. This configuration effectively captures both local and global features, thereby enhancing the model's ability to represent complex EEG data. The dual-branch architecture, along with multi-scale feature fusion, utilizes the original noisy EEG signal as a conditioning factor to guide the denoising process. The local information unit (CNN structure) extracts local features, while the Transformer encoder block models global dependencies. A fusion module integrates these

features, adjusts channel counts, and produces the denoised EEG signal. Additionally, we introduce a feature linear modulation (FiLM) [38] module to incorporate prior knowledge into the denoising model, accelerating convergence and optimizing training.

The dual-branch denoising network used to fit the denoising function ϵ_θ . During training, as illustrated in Fig. 3, the dual-branch network takes the original noisy EEG signal $\tilde{x}$ and the signal x from the noise sequence as inputs. Each branch first processes the signals through a local information unit to extract

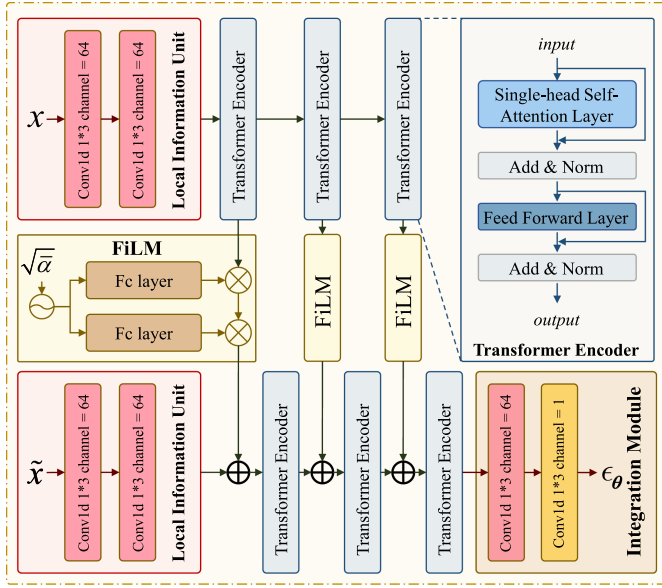


Fig. 3. Denoising model in diffusion model. The figure provides the overall framework of the denoising model and details the structure of each module.

initial local features, followed by three Transformer encoder layers to capture global dependencies. The FiLM module maps the noise level $\sqrt{\alpha}$ to the feature scale and completes multi-scale feature fusion across the two branches [39]. At the end of the model, an integration module maps these features to the dimension of the target signal. Detailed rparameters for each module will be provided in the subsequent experimental setting section.

IV. EXPERIMENTS

This section outlines the experimental setting for assessing the proposed denoising model. It begins with an introduction to the dataset utilized in the experiments and the preprocessing techniques applied to ready the data for training and testing the model. Following this, it details the experimental configurations, including hyperparameter settings and metrics for evaluating the model. Finally, it presents the results of the experiments and a comparative analysis of the proposed model.

A. Dataset and Data Preprocessing

The following section describes the organization of each dataset and details the preprocessing methods used for each dataset.

1) *EEGdenoiseNet Dataset*: The EEGdenoiseNet dataset comprises 4,514 EEG signal segments, 3,400 EOG signal segments, and 5,598 EMG signal segments. Each segment is 2 seconds in duration and has been standardized. The EOG signals are sampled at 256 Hz, while the EMG signals are sampled at 512 Hz. The EEG data are resampled to either 256 Hz or 512 Hz to incorporate EOG and EMG noise. Datasets with SNRs ranging from -5 dB to 5 dB were created, and the data is split into training, validation, and test sets in an 8:1:1 ratio [12].

2) *Semi-Simulated EEG/EOG Dataset (SSED)*: The SSED dataset consists of 54 30-second multi-channel EEG recordings from 27 subjects. Each recording has 19 EEG channels sampled at 200Hz. EOG data was also collected from the same subjects, with recordings of vertical and horizontal EOG used to synthesize contaminated EEG signals. To facilitate model training, the data signals were standardized due to their large amplitude. To satisfy the data volume requirements for training deep models, we divided the multi-channel EEG data into single-channel segments. During the inference phase, each channel is denoised independently. The dataset was divided into training, validation, and testing sets with an 8:1:1 ratio [40].

3) *MNE Sample Dataset*: The MNE Sample Dataset is extensively used for analyzing EEG. This dataset comprises EEG data from 60 electrodes, including 59 EEG channels and 1 EOG channel, capturing brain activity during visual and auditory tasks. During the experiments, subjects experienced visual stimuli via a checkerboard pattern randomly presented in both visual fields every 750 milliseconds, and auditory stimuli through monaural tones in each ear [41]. This dataset is instrumental in validating EEG denoising methods.

4) *Resting State EEG Data*: The dataset comprises continuous EEG signals recorded from 72 channels [42], including electrodes placed under the eyes. The signals were amplified using the BioSemi Active II system in 24-bit DC mode with an initial sampling rate of 2,048 Hz (400 Hz bandwidth) and downsampled online to 256 Hz. Data collection involved 22 undergraduate students. Participants sat quietly in a dark room on a cushioned chair for 8 minutes of resting-state EEG recording—4 minutes with eyes open and 4 minutes with eyes closed, alternating every 1 minute [43].

5) *Motion Artifact Contaminated EEG Data*: The experimental data were obtained from six healthy subjects across a total of 24 trials. EEG channels 1 and 2 were recorded from the frontal cortex at positions FPz and FP1h, respectively, following the 10-5 system. During the experiment, subjects were instructed to keep their eyes closed and maintain a steady head position to minimize artifacts arising from blinking, saccades, and head movements. Each trial lasted for 9 minutes, with motion artifacts mechanically induced by pulling the connecting lead on the electrode of channel 2 at 2-minute intervals [44].

B. Evaluation Metrics

Five evaluation metrics were selected to measure the performance of the models, including the relative root mean square error in the temporal domain ($\text{RRMSE}_{\text{temporal}}$), the relative root mean square error in the spectral domain ($\text{RRMSE}_{\text{spectral}}$), the correlation coefficient (CC), and the p -value.

- $\text{RRMSE}_{\text{temporal}}$: This metric quantifies the discrepancy between the denoised signal and the clean signal in the temporal domain. It is defined as:

$$\text{RRMSE}_{\text{temporal}} = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}}{\sqrt{\frac{1}{N} \sum_{i=1}^N (y_i)^2}}, \quad (14)$$

where N represents the total number of time-domain sampling points, y_i denotes the i -th sampling point of the clean signal, and $\hat{y}_i$ represents the corresponding i -th sampling point in the denoised sample.

- $\text{RRMSE}_{\text{spectral}}$: This metric evaluates the difference between the denoised signal and the clean signal in the spectral domain. It is defined as:

$$\text{RRMSE}_{\text{spectral}} = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2}}{\sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i)^2}}, \quad (15)$$

where Y_i and $\hat{Y}_i$ represent the power spectral density (PSD) values of the clean EEG signal and the corresponding denoised EEG signal at frequency bin i , respectively. N represents the total number of frequency bins in the PSD.

- CC : This metric quantifies the linear relationship between the denoised signal and the clean signal. It is calculated as follows:

$$\text{CC} = \frac{\sum_{i=1}^N (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^N (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})^2}} \quad (16)$$

where $\bar{y}$ and $\bar{\hat{y}}$ are the means of the clean and denoised signals, respectively.

- p -value: This metric assesses the significance of the correlation coefficient. Typically, a p -value below 0.05 is regarded as significant.

C. Experimental Setting

In this section, we outline the experimental setting and implementation details. We set the diffusion steps to 500 and the range of β to [0.0001, 0.02]. For the Transformer model, we set the dimensions of Q, K, and V to 64. For the SSED datasets with smaller data volumes, we adjusted the dimensions to 32. To accommodate the limited quantity of EEG samples, the number of heads in the original multiple self-attention layers of the Transformer Encoder module was adjusted to 1. During training, the batch size was set at 512, and the model was trained across 4000 epochs. We initially set the learning rate at 0.001 and decreased it by a factor of 10 every 1500 epochs. The model parameters were optimized using the Adam optimizer. All programming was done using Python 3.9 and PyTorch 1.12.0, with experiments conducted on an Nvidia GeForce RTX 4070 ti GPU. The code is available at <https://github.com/XYH0118/EEGDFUS.git>.

The tendency of the model to overfit when using a generative learning strategy is a significant challenge. To mitigate overfitting during training, we employ a validation set to monitor model performance. Specifically, we calculate the model's loss on the validation set every 20 epochs. If the validation loss is the lowest recorded, the model is saved; otherwise, training continues for the next round. By monitoring performance through the validation set, we can select the best-performing model, thereby achieving optimal denoising results on the test set.

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON SSED

Method	RRMSE _{temporal}	RRMSE _{spectral}	CC
filter [45]	0.772	2.011	0.764
fcNN [12]	0.532	0.648	0.880
simple_CNN [12]	0.481	0.721	0.894
complex_CNN [12]	0.445	0.701	0.910
RNN.LSTM [12]	0.613	1.015	0.862
GAN [18]	0.274	0.277	0.965
Transformer [46]	0.222	0.202	0.975
EEGDFUS	0.121	0.127	0.992

• All CC values are accompanied by p -values lower than 0.05.

D. Overall Performance

Firstly, we present the experimental results on the EEGdenoiseNet dataset, which are detailed in Fig. 4 for different input SNR levels. This figure illustrates the performance of different methods, including our proposed EEGDFUS and baseline models [12], [18], [45], [46]. It is evident from the results that EEGDFUS consistently outperforms all baseline methods in every evaluation metric, showcasing significant improvements over the existing approaches. Additionally, we provide an overall performance comparison of the models on the SSED dataset, which is presented in Table I. Similarly, the EEGDFUS achieves the best performance, with a notable improvement over the baseline methods. The experimental results presented in this work demonstrate that the proposed EEGDFUS is a highly effective denoising model. It significantly reduces noise while preserving the active components of the original signal to a remarkable degree. These findings highlight the potential of the EEGDFUS for practical applications in signal denoising, where the accurate preservation of important signal features is crucial for downstream analysis and interpretation.

To visually illustrate the effectiveness of our proposed method, we selected some samples of contaminated EEG signals from the SSED test dataset and compared the signals before and after applying our denoising approach, as shown in Fig. 5. Our results demonstrate that even when the noise amplitude is high, our EEGDFUS can effectively reduce noise and preserve the active components of the original signal, resulting in high-quality EEG signals. Furthermore, even in cases where the noise amplitude is very small, and the contaminated EEG signal is similar to the clean EEG signal, our EEGDFUS model can still maintain the signal's integrity and retain the active components without affecting the information contained in the signal. The experimental results indicating its effectiveness in denoising EEG signals under different noise conditions.

The application of the diffusion model for EEG denoising involves the progressive restoration of clean EEG signals from standard Gaussian noise. To illustrate this process, we present the corresponding denoising process of one of the contaminated EEG signals in Fig. 6. We randomly selected signals from the test set of the EEGdenoiseNet dataset, which serves as additional evidence of the proposed EEGDFUS's exceptional performance on this dataset. The total number of diffusion steps is 500, and we show the diffusion state every 50 steps. Subgraph (a) represents the standard Gaussian noise as the input to the model,

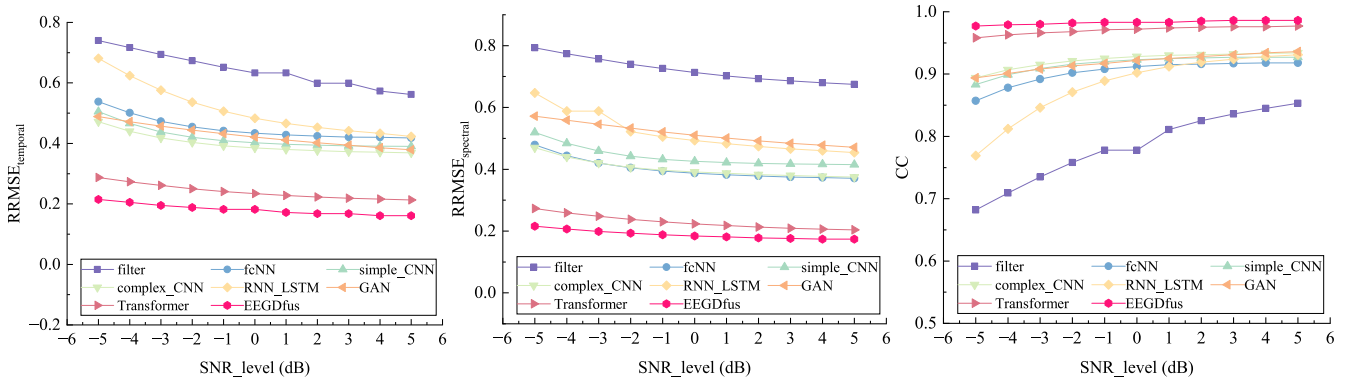


Fig. 4. Performance of different models in removing ocular artifact at various SNR levels on the EEGdenoiseNet dataset.

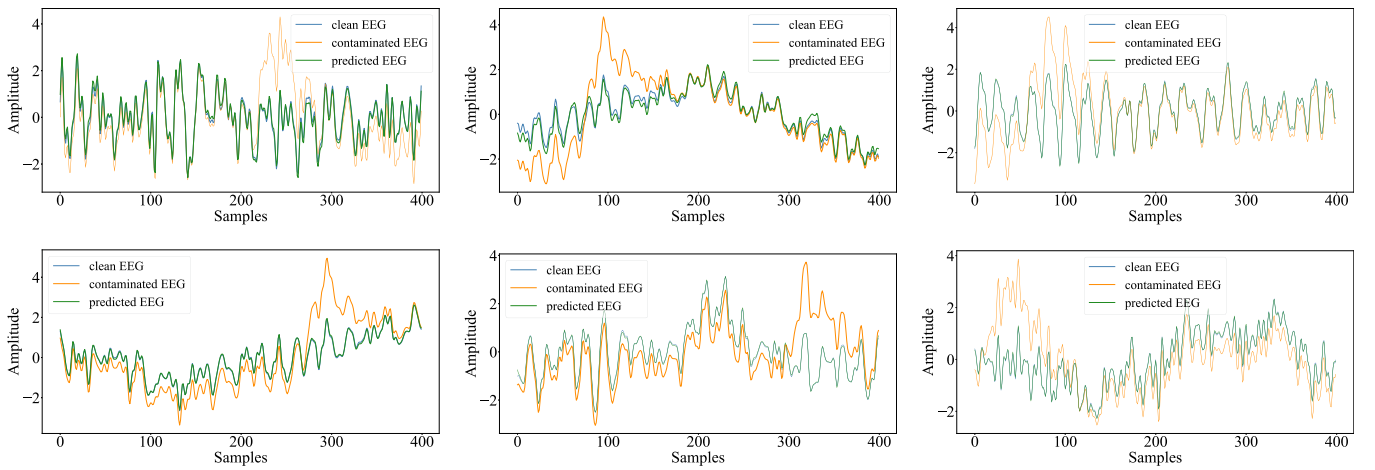


Fig. 5. Comparison of contaminated EEG, denoised EEG, and clean EEG signals.

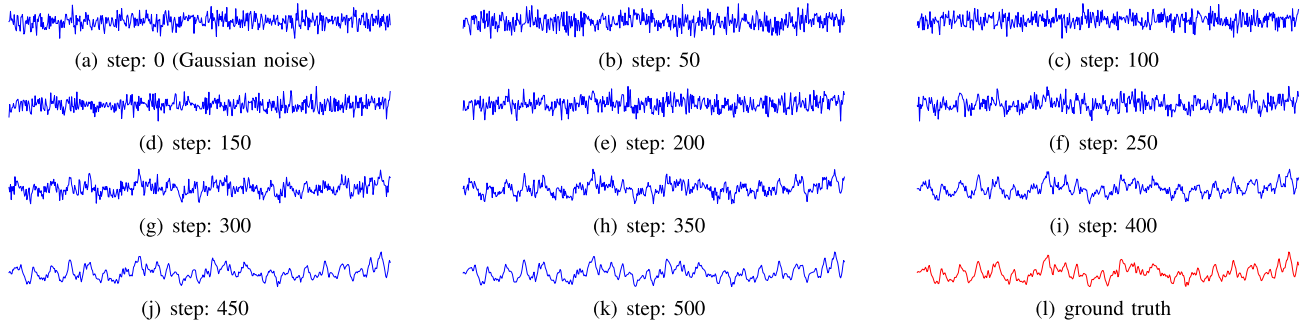


Fig. 6. Denoising process of EEGDfus. The total iteration is 500 steps, and the denoising performance is displayed every 50 steps.

while subgraph (b) represents the contaminated EEG signal as the condition. Subgraphs (c) through (m) illustrate the entire diffusion process, while subgraph (n) represents the clean EEG signal as a label.

As shown in the figure, the model iteratively denoises the standard Gaussian signal as the number of diffusion steps increases, gradually restoring the clean EEG signal. The closer the diffusion process is to completion, the finer and higher the quality of the generated signal. Ultimately, the model generates

a denoised EEG signal that is very similar to the clean EEG label, completing the denoising process of the EEG.

E. EEGDfus Denoising Performance for Other Types of Artifacts

To verify the applicability of EEGDfus to other types of artifacts, we conducted supplementary experiments focusing on muscular and motion artifacts. These experiments evaluated the

TABLE II

EEGDFUS DENOISING PERFORMANCE FOR MUSCULAR ARTIFACTS

Method	RRMSE _{temporal}	RRMSE _{spectral}	CC
fcNN [47]	0.448	0.451	0.898
SCNN [12]	0.430	0.401	0.895
NovelCNN [12]	0.261	0.274	0.958
SDNet [48]	0.424	0.397	0.898
EEGIFNet [49]	0.321	0.321	0.946
GCTNet [50]	0.241	0.298	0.972
EEGDFus	0.171	0.154	0.989

TABLE III

EEGDFUS DENOISING PERFORMANCE FOR MOTION ARTIFACTS

Method	RRMSE _{temporal}	RRMSE _{spectral}	CC
fcNN [47]	0.509	0.605	0.873
SCNN [12]	0.381	0.365	0.926
NovelCNN [12]	0.373	0.349	0.931
SDNet [48]	0.364	0.333	0.935
EEGIFNet [49]	0.347	0.303	0.945
GCTNet [50]	0.171	0.173	0.991
EEGDFus	0.149	0.156	0.991

effectiveness of EEGDFus in handling these artifacts, with the results summarized in Tables II and III.

Compared to ocular artifacts, muscular artifacts are more complex and exhibit a higher degree of overlap with EEG signals. Therefore, we established a new set of model parameters specifically for muscular artifacts to better extract EMG and EEG features. We specifically configured the following hyper-parameters: the number of convolutional channels ($hidden_{dim}$) was set to 128, the number of heads in the Transformer Encoder ($head$) was set to 2, the dimension of QKV (qkv_{dim}) was set to 64, and the hidden dimension of the fully connected layer in the Transformer Encoder was set to 2048. The number of diffusion steps and training epochs remained unchanged. This approach significantly increased the complexity of the denoising model, enabling it to better extract noise features. The model parameters for removing motion artifacts are consistent with those used for removing ocular artifacts. The experimental results show that EEGDFus matches the performance of state-of-the-art methods in addressing muscular and motion artifacts. This highlights the advanced denoising capabilities and robust generalization of EEGDFus, affirming its effectiveness across diverse and challenging EEG denoising scenarios.

F. Real-World EEG Denoising

We use the MNE sample dataset to evaluate the denoising performance of our model on real world noisy EEG. In Fig. 7, we present a comparison of the denoising effects of EEGIFNet, NovelCNN, and EEGDFus on real-world noisy EEG data. Subfigures (a) and (b) are derived from different EEG channels during the same time period, while subfigures (c) and (d) originate from the same EEG channel across different time periods. The results indicate that EEGDFus surpasses EEGIFNet in effectively eliminating ocular artifacts, achieving superior denoising performance. Although NovelCNN more thoroughly removes ocular artifacts, it significantly compromises critical EEG signal information, resulting in an inability to achieve refined denoised EEG signals. In contrast, EEGDFus better

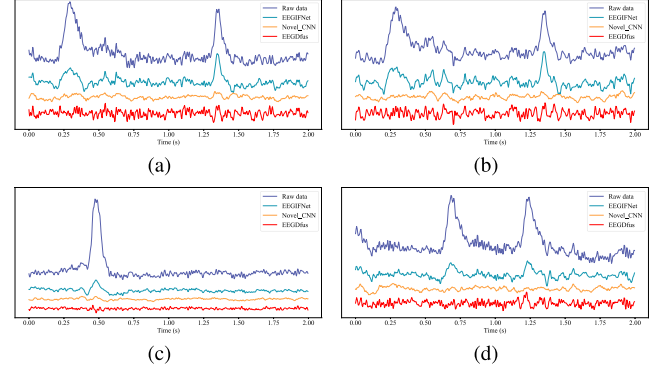


Fig. 7. Comparison of the denoising performance of different methods on real-world noisy EEG signals.

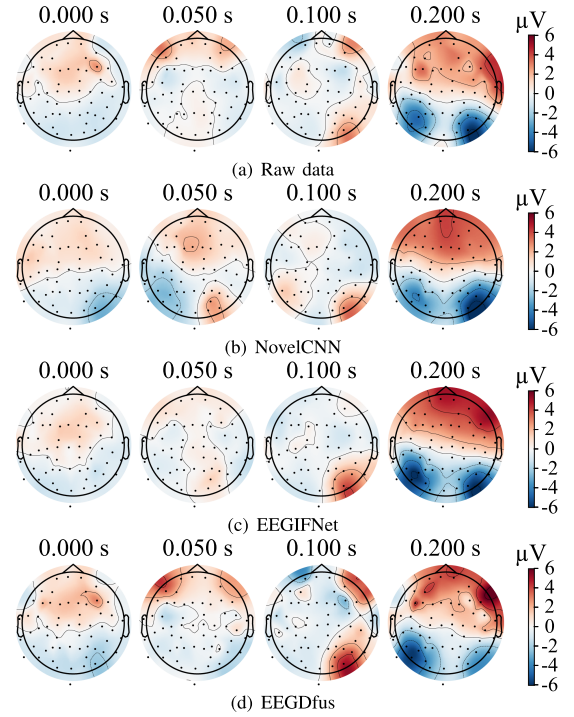


Fig. 8. Topographic mapping of EEG activity. This figure presents topographic maps of Event-Related Potentials (ERPs) derived from different EEG signals. The maps correspond to ERPs from the raw EEG signal, ERPs from the EEG signal denoised by NovelCNN, ERPs from the EEG signal denoised by EEGIFNet, and ERPs from the EEG signal denoised by EEGDFus at 0 s, 0.05 s, 0.1 s, and 0.2 s.

preserves signal details compared to NovelCNN, achieving more precise EEG denoising.

In addition, we analyze event-related potential (ERP) to determine whether the proposed denoising method can retain useful information in the EEG signal while removing noise. We select the first 100 milliseconds and the last 400 milliseconds of the right visual field stimulus and average multiple trials to obtain the ERP [51]. The denoising model is then applied to remove ocular artifacts. Fig. 8 display the topography of the raw EEG signal, the EEG signal denoised by NovelCNN, the EEG signal denoised by EEGIFNet, and the EEG signal denoised by EEGDFus.

TABLE IV
SYNCHRONICITY ANALYSIS OF SYNCHRONOUS SIGNALS BEFORE AND AFTER DENOISING

EEG samples	CC-BD	CC-AD	PLV-BD	PLV-AD
sample_0	0.229	0.900	0.084	0.785
sample_1	0.311	0.838	0.389	0.713
sample_2	0.762	0.925	0.723	0.848
sample_3	0.917	0.914	0.822	0.805
sample_4	0.628	0.862	0.705	0.764
sample_5	0.843	0.892	0.791	0.802
sample_6	0.877	0.910	0.805	0.843
sample_7	0.958	0.900	0.926	0.831
sample_8	0.953	0.894	0.922	0.810
sample_9	0.954	0.904	0.930	0.838
Average	0.743	0.894	0.710	0.804

TABLE V
SYNCHRONICITY ANALYSIS OF NON-SYNCHRONOUS SIGNALS BEFORE AND AFTER DENOISING

EEG samples	CC-BD	CC-AD	PLV-BD	PLV-AD
sample_0	0.403	-0.122	0.278	0.102
sample_1	-0.120	0.173	0.214	0.160
sample_2	0.346	0.226	0.341	0.222
sample_3	-0.187	0.125	0.249	0.182
sample_4	-0.296	0.075	0.556	0.146
sample_5	-0.273	0.012	0.331	0.095
sample_6	0.009	-0.199	0.336	0.191
sample_7	-0.484	-0.256	0.460	0.316
sample_8	-0.399	-0.230	0.381	0.268
sample_9	-0.163	-0.037	0.343	0.112
Average	-0.116	-0.023	0.349	0.179

Early sensory components typically appear around 100 milliseconds after stimulus presentation [52], [53]. Visual stimuli to the left eye are generally reflected as activations in the right occipital lobe of the brain. As illustrated in Fig. 8, the EEG signals, after being denoised using the EEGDfus method, show significant activation in the right occipital region of the brain. This is more pronounced compared to other methods. This finding confirms that the proposed EEGDfus method can effectively retain useful components within the EEG signals. Integrating the results from both experiments, it is evident that EEGDfus achieves high-precision EEG denoising while preserving essential information within the EEG signals, even in the presence of real noise.

G. Multi-Channel EEG Denoising Performance

To verify whether the proposed EEGDfus method can capture synchronization between EEG signal channels, we conducted experiments using multi-channel noisy EEG from the SSED dataset. We randomly selected ten pairs of synchronized EEG signals (signals from two adjacent channels recorded during the same time period) and ten pairs of non-synchronized EEG signals (signals from two adjacent channels recorded during different time periods). We then computed their correlation coefficients (CC) and phase-locking values (PLV) both before and after denoising to assess phase synchronization within specific frequency bands. The PLV ranges from 0 to 1, where a value of 1 signifies complete synchronization and 0 signifies complete desynchronization. The findings are summarized in Tables IV and V.

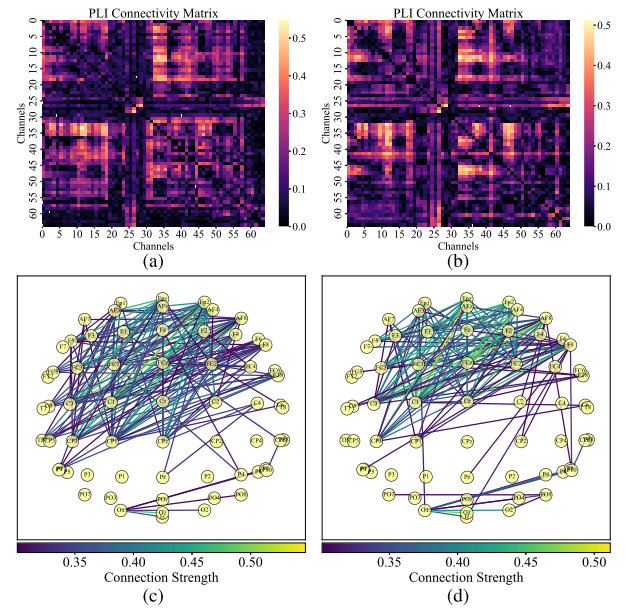


Fig. 9. Resting-state brain network. Subplots (a) and (b) show heatmaps of the PLI matrix for the multichannel EEG signals before and after denoising, respectively. Subplots (c) and (d) depict the brain functional connectivity strength networks corresponding to subplots (a) and (b), respectively.

For synchronized EEG signals, the correlation coefficient and phase-locking value of the two channels significantly improved after denoising. The average CC increased from 0.745 to 0.894, and the average PLV increased from 0.710 to 0.804. This demonstrates that, in most cases, EEGDfus effectively maintains synchronization in EEG samples after denoising. Despite this, the overall synchronization of denoised EEG signals remains high. For non-synchronized samples, the synchronization and correlation of EEG signals remain low after denoising, validating the proposed verification method. The above experiments demonstrate that EEGDfus can be effectively applied to multi-channel EEG denoising without disrupting the synchronization between channels.

H. Clean EEG Information Retention Performance

To validate the denoising performance of the proposed EEGDfus model on real multichannel EEG signals, we conducted experiments using the Resting State EEG Data dataset [43]. This dataset contains scalp EEG recordings from 22 participants, each with 72 channels, captured while the participants were in an awake resting state. Materials and Experiments. For our experiments, we selected 64 channels that adhere to the international 10-10 electrode placement system. In line with previous studies, we employed the Phase Lag Index (PLI) to measure functional brain connectivity, with the PLI matrix reflecting the resting-state brain network [54]. Fig. 9 presents a heatmap of the PLI matrix, comparing pre-denoising and post-denoising EEG signals, alongside the corresponding brain network connectivity graphs. In these graphs, connections with a PLI greater than 0.3 are shown.

TABLE VI

COMPARATIVE PERFORMANCE ANALYSIS OF EXISTING DENOISING MODELS ON THE EEGDENOISENET DATASET

Method	RRMSE _{temporal}	RRMSE _{spectral}	CC
RNN_LSTM [12]	0.740	0.696	0.677
DLN [55]	0.699	0.579	0.720
SCNN [12]	0.620	0.526	0.791
1D-ResCNN [13]	0.630	0.588	0.776
EEGDnet [14]	0.497	0.491	0.868
FIEM_LFCNN [56]	0.350	-	0.909
FIEM_SimpleCNN [56]	0.329	-	0.922
FIEM_ResCNN [56]	0.331	-	0.919
SDNet [48]	0.267	0.272	0.956
EEGIFNet [49]	0.206	0.249	0.976
GCTNet [50]	0.184	0.193	0.981
EEGDFus	0.182	0.188	0.983

TABLE VII

COMPARATIVE PERFORMANCE ANALYSIS OF EXISTING DENOISING MODELS ON THE SSED DATASET

Method	RRMSE _{temporal}	RRMSE _{spectral}	CC
fcNN [47]	0.459	0.414	0.899
LSTM [47]	0.419	0.413	0.920
1D-ResCNN [13]	0.336	0.371	0.941
NovelCNN [57]	0.260	0.269	0.965
DuoCL [47]	0.217	0.161	0.976
EEGDFus	0.121	0.127	0.992

The figure demonstrates that denoising resting-state EEG using EEGDFus does not disrupt the inherent resting-state brain network. The PLI matrix remains largely consistent before and after denoising, with the main functional connections nearly identical. This consistency validates that EEGDFus can accurately identify noise characteristics without compromising essential information in the EEG signals. Additionally, for multichannel EEG, the model effectively preserves inter-channel connectivity, maintaining the original, valuable features of the signals.

V. DISCUSSION

A. Comparison of our Approach With Current State-of-the-art Solutions

In this paper, we introduce EEGDFus, a conditional diffusion model for fine-grained EEG denoising. We compare EEGDFus with current state-of-the-art methods, with detailed results shown in Tables VI and VII. The experimental findings indicate that EEGDFus surpasses other methods in performance on the two public datasets.

On the EEGdenoiseNet dataset, our proposed EEGDFus achieves the lowest RRMSE_{temporal} and RRMSE_{spectral}, and the highest CC, and on the SSED dataset, the proposed EEGDFus also achieves the lowest RRMSE_{temporal} and RRMSE_{spectral}, and the highest CC. A low RRMSE_{temporal} and RRMSE_{spectral} imply that the generated signal is similar to the ground truth in both the time and frequency domains; and a high CC means that the generated post-denoised EEG is highly correlated with the clean label EEG. Specifically, our proposed EEGDFus achieves RRMSE_{temporal}, RRMSE_{spectral}, and CC of 0.182, 0.188, and 0.983, respectively, on EEGdenoiseNet, and on the SSED dataset, our proposed EEGDFus achieves 0.121 RRMSE_{temporal}, 0.127 RRMSE_{spectral}, and 0.992 CC. These results are optimal performance on both datasets.

TABLE VIII

GENERALIZATION PERFORMANCE OF EXISTING DENOISING MODELS

Method	RRMSE _{temporal}	RRMSE _{spectral}	CC
fcNN [47]	0.769	0.774	0.634
SCNN [12]	0.646	0.612	0.740
NovelCNN [12]	0.794	0.711	0.587
SDNet [48]	0.713	0.655	0.679
GCTNet [50]	0.716	0.796	0.710
EEGIFNet [49]	0.576	0.600	0.792
EEGDFus	0.556	0.571	0.806
EEGDFus_finetune	0.391	0.485	0.925

TABLE IX

IMPACT OF HYPERPARAMETERS ON MODEL PERFORMANCE

Method	RRMSE _{temporal}	RRMSE _{spectral}	CC
hidden_dim_32	0.135	0.125	0.990
hidden_dim_64	0.121	0.127	0.992
hidden_dim_128	0.183	0.174	0.982
head_1	0.121	0.127	0.992
head_4	0.371	0.573	0.967
head_8	0.989	0.999	0.872
qkv_dim_16	0.967	0.999	0.959
qkv_dim_32	0.121	0.127	0.992
qkv_dim_64	0.363	0.572	0.964

B. Generalization Effectiveness Analysis

To evaluate the generalization performance of the proposed EEGDFus, we initially trained the model on the EEGdenoiseNet dataset. We then performed quantitative tests using the SSED dataset. Furthermore, we applied this same approach to assess the generalization performance of various baseline models and the most recent denoising models. The experimental outcomes are summarized in Table VIII.

The experimental results show that the generalization performance of current data-driven deep denoising methods is lacking due to substantial differences between datasets. In contrast, the proposed EEGDFus achieves relatively superior generalization performance. This demonstrates that EEGDFus can model EEG features more accurately than existing methods, resulting in better generalization. We further employed a transfer learning method, using 10% of the data from the SSED dataset to fine-tune the trained model. The results after fine-tuning, represented as EEGDFus_finetune in the table, reveal that even with a small amount of data, the model's denoising performance improved significantly.

C. Hyperparameter Adjustment Strategies

This section primarily focuses on the hyperparameter settings of the model and presents experiments conducted using the SSED dataset. The key hyperparameters of the proposed dual-branch denoising model include the number of convolution channels in the local information unit, denoted as *hidden_dim*, the number of heads in the Transformer encoder's multi-head self-attention mechanism, referred to as *head*, and the dimension of the multi-head self-attention mechanism's QKV, labeled as *qkv_dim*. We conducted supplementary experiments on these hyperparameters, with the results presented in Table IX.

The results of the experiments reveal that the model performs best on the SSED dataset with the hyperparameters configured as *hidden_dim* = 64, *head* = 1, and *qkv_dim* = 32. It is

TABLE X
PERFORMANCE AND INFERENCE TIME ACROSS VARIOUS NOISE SEQUENCES AND SAMPLING STEPS

Noise schedule	Steps	Inference time (ms)	RRMSE _{temporal}	RRMSE _{spectral}	CC
linear	50	3.892	0.143	0.134	0.989
quadratic	50	3.198	0.155	0.176	0.988
sigmoid	50	3.209	0.181	0.204	0.984
polynomial	50	3.862	0.173	0.225	0.985
exponential	200	15.105	0.133	0.124	0.990
linear (default)	500	85.365	0.121	0.127	0.992

worth noting that the number of heads and QKV values are core parameters of the attention mechanism. Adjusting these parameters simultaneously affects the performance of the six Transformer Encoder modules, making the model sensitive to these parameters. Insufficient parameters result in inadequate model fitting, while excessive parameters cause overfitting. The table data indicates that when the number of heads is too high (8) and the number of QKVs is too low (16), the model loses its ability to effectively model the data. The appropriate number of parameters can enable the model to achieve the best performance.

D. Karras Noise Schedule Analysis

In this section, we further investigated the Karras noise schedule [58], which is a technique used in diffusion models, particularly focusing on improving the noise schedule – the plan for gradually reducing noise during the generation process. The core idea of the Karras method is to reduce the noise step size in the later stages of sampling, which in turn improves the quality of the generated output.

We performed experiments with the SSED dataset to assess the impact of different noise scheduling techniques and varying numbers of diffusion steps on both model inference time and denoising efficacy. The results are summarized in Table X. The table lists the average time required for the model to denoise a noisy EEG signal with a two-second duration and a 200 Hz sampling rate. Models with an inference time under 2 seconds are deemed suitable for real-time applications.

In the table, the best performance and shortest time are highlighted in blue, while the worst performance and longest time are marked in red. The results demonstrate that even with fewer diffusion steps, the model can still achieve satisfactory performance and reduce inference time, thus meeting the diverse and complex requirements for EEG denoising. The model's inference time is significantly less than two seconds, well-suited to meet real-time demands.

E. Advantages and Limitations of EEGDfus

The proposed EEGDfus system excels in EEG denoising by leveraging the diffusion model's capacity to accurately learn the distribution of clean EEG signals. We introduced an innovative dual-branch denoising approach that integrates the original noisy EEG signal as a condition to guide the generation of the cleaned signal. This model combines the strengths of Transformers for capturing global information and local information units for extracting essential features to remove artifacts effectively. Additionally, the FiLM model's incorporation of noise levels

aids in model convergence and enhances the control of conditional information by fusing multi-scale features. EEGDfus has shown promising performance on public datasets. However, like other data-driven deep learning methods, it faces challenges with generalization. Future research will focus on enhancing the generalization capabilities of EEG denoising models.

VI. CONCLUSION

In this paper, we present EEGDfus, an innovative conditional diffusion model specifically designed for fine-grained EEG denoising. Compared to traditional autoregressive methods, EEGDfus better preserves the high-frequency information of the predicted signal and is more stable during training. Experimental results demonstrate that EEGDfus excels in fine-grained EEG denoising, surpassing current state-of-the-art methods across multiple public datasets. Specifically, for EOG artifact denoising, EEGDfus achieved a temporal RRMSE of 0.182, a spectral RRMSE of 0.188, and a correlation coefficient (CC) of 0.983 on the EEGdenoiseNet dataset. Additionally, on the SSED dataset, it achieved a temporal RRMSE of 0.121, a spectral RRMSE of 0.127, and a CC of 0.992. EEGDfus also achieves satisfactory performance in reducing other noises and excels in real-world noisy EEG denoising tasks. These results underscore the effectiveness and superiority of EEGDfus in artifact removal, establishing the diffusion model as a valuable tool for high-performance EEG denoising. This makes EEGDfus an effective solution for enhancing subsequent EEG analysis and interpretation.

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